

JAY JOSHI

+91 8875549960 joshijayy421@gmail.com LinkedIn GitHub Portfolio

Education

Indian Institute of Information Technology

Bachelor of Technology in Computer Science

August 2022 – July 2026

Kota, Rajasthan

Swami Vivekananda Government Model School

Class XII CBSE (Science Stream)

March 2021 – July 2022

Reodar, Rajasthan

Experience

Freelance AI & Software Developer

Remote

Freelancer.com

- Built and deployed Python-based AI and data solutions for global clients across finance and AI research domains, achieving a **4.8-star rating**, **13+ verified reviews**, and **100% job success rate** on Freelancer.com.
- Engineered a Swing Reversion Grid algorithmic trading strategy in Python using a **50-period SMA** as the mean anchor, with multi-level grid entries at fixed percentage steps (**-2%, -4%, -6% below SMA**), mean-reversion exits, and a **15% stop-loss** — ranked 1st globally in a Freelancer trading strategy contest, winning **\$1,000 USD**.
- Built Python service using the **j-hartmann/emotion-english-distilroberta-base transformer model** for emotion classification from text, mapping detected emotions to TTS voice parameters (**rate, volume, mood-based voice selection**) to generate modulated WAV audio output, with a Flask web UI and CLI interface.

Projects

DevDoc AI | *Python, LangChain, FastAPI, Next.js, HuggingFace Embeddings, Vector Search, OAuth* 🔒 ↗

- Built a full-stack RAG platform that indexes GitHub repos into a vector store via a **5-stage pipeline (clone → chunk → embed → retrieve → generate)** using LangChain and sentence-transformer embeddings, returning answers with source citations.
- Supports **3 indexing modes** (full, incremental, file-level), **2 query modes** (single/multi-repo), and **multi-provider LLM routing** across Groq, Gemini, and HuggingFace with automatic fallback.
- Designed **10+ REST API** endpoints covering Q&A, repo insights, doc generation, and export — with safety limits (**250 files, 200 KB/file, 1.2M chars**) deployed live on Render + Vercel with GitHub + Google OAuth.

Rythm AI | *Machine Learning, Feature Engineering, PCA, Data Preprocessing, EDA, Model Evaluation* 🔒

- Built a **16-class cardiac arrhythmia classifier** on UCI dataset (**452 patients, 279 features**) — evaluated 7 algorithms; best model Kernelized SVM with PCA achieved **80.21%** accuracy on test data.
- Applied PCA for dimensionality reduction on **279 features and SMOTE oversampling** to handle severe class imbalance across **12 arrhythmia types**, improving model stability significantly.
- Ran full **EDA + Preprocessing pipeline** covering missing value imputation, feature scaling, and class distribution analysis — compared all models with/without PCA using **train vs. test accuracy charts**.

StockFlow | *MySQL Workbench, SQL (Joins, Aggregations), Python, Jupyter Notebook* 🔒

- Designed a normalized relational database schema with **6 interconnected tables** (Products, Categories, Warehouses, Inventory, Customers, Orders) with full foreign key constraints.
- Built an ETL pipeline using Pandas and SQLAlchemy to bulk-load data from **6 CSV files into MySQL** — automated table creation, data cleaning, and insertion in a single notebook run.
- Wrote SQL analysis scripts (**joins, aggregations, stored procedures**) to query inventory levels, stock movements, and order fulfillment across **101 warehouses and 1,000+ orders**.

Technical Skills

Languages, Frameworks & Databases: Python, JavaScript, TypeScript, SQL, C, C++, FastAPI, Next.js, React, Node.js, Express.js, Streamlit, MySQL, MongoDB

AI / GenAI / RAG: LangChain, RAG, HuggingFace Transformers, Sentence Transformers, Gemini API, Groq API, OpenAI API, Vector Search

ML / Data: Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, PCA, SMOTE, EDA

Dev Tools & Deployment: Git / GitHub, Docker, VS Code, Linux, GitHub Actions, Vercel, Render

Leadership / Extracurricular

- Chief Video Editor at Paryavaran Shakti and ARC Robotics Club, IIIT Kota — led content production and post-production for both clubs simultaneously.
- Co-organized a college-level hackathon under Codebase Club, IIIT Kota — managed end-to-end logistics, participant coordination, and event execution.
- Led a team at ReImagine Hackathon by Sheryians (June 2024, Bhopal) — delivered a redesigned responsive website under contest deadline.